

WEEK 8: CREATE A DATA VISUALIZATION (E.G., PIE CHARTS, BAR GRAPHS) FOR AN INVENTORY MANAGEMENT SYSTEM USING JAVASCRIPT

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AIM: The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File First, create an HTML file to hold your canvas for the chart and include Chart.js.

```
html
□<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Inventory Management Visualization</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin: 50px;
    }
    canvas {
      margin: 20px auto;
    }
  </style>
</head>
<body>
```

```

    </style>
</head>
<body>
    <h1>Inventory Management System</h1>
    <canvas id="pieChart" width="400" height="400"></canvas>
    <canvas id="barChart" width="400" height="400"></canvas>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <script src="script.js"></script>
</body>
</html>

```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

javascript

□// script.js

// Data for the inventory

```

const inventoryData = {
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
  datasets: [
    {
      label: 'Items in Stock',
      data: [200, 150, 100, 80, 50],
      backgroundColor: [
        '#FF6384',
        '#36A2EB',
        '#FFCE56',
        '#4BC0C0',

```

```
        '#9966FF'
    ],
  }
]
};
```

// Creating the Pie Chart

```
const ctxPie = document.getElementById('pieChart').getContext('2d');
const pieChart = new Chart(ctxPie, {
  type: 'pie',
  data: inventoryData,
  options: {
    responsive: true,
    title: {
      display: true,
      text: 'Inventory Distribution'
    }
  }
});
```

// Creating the Bar Chart

```
const ctxBar = document.getElementById('barChart').getContext('2d');
const barChart = new Chart(ctxBar, {
  type: 'bar',
  data: inventoryData,
  options: {
    responsive: true,
    title: {
      display: true,
```

```

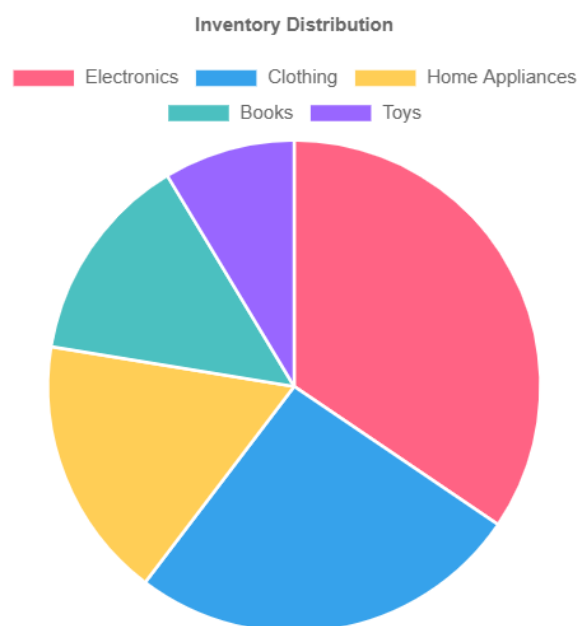
        text: 'Items in Stock by Category'
    },
    scales: {
        yAxes: [{
            ticks: {
                beginAtZero: true
            }
        }]
    }
}
});

```

OUTPUT:

1. Inventory Distribution – Pie Char:

Inventory Management System



The pie chart visually represents the proportional distribution of inventory items across various categories. It is a high-level summary used for understanding the overall stock makeup at a glance.

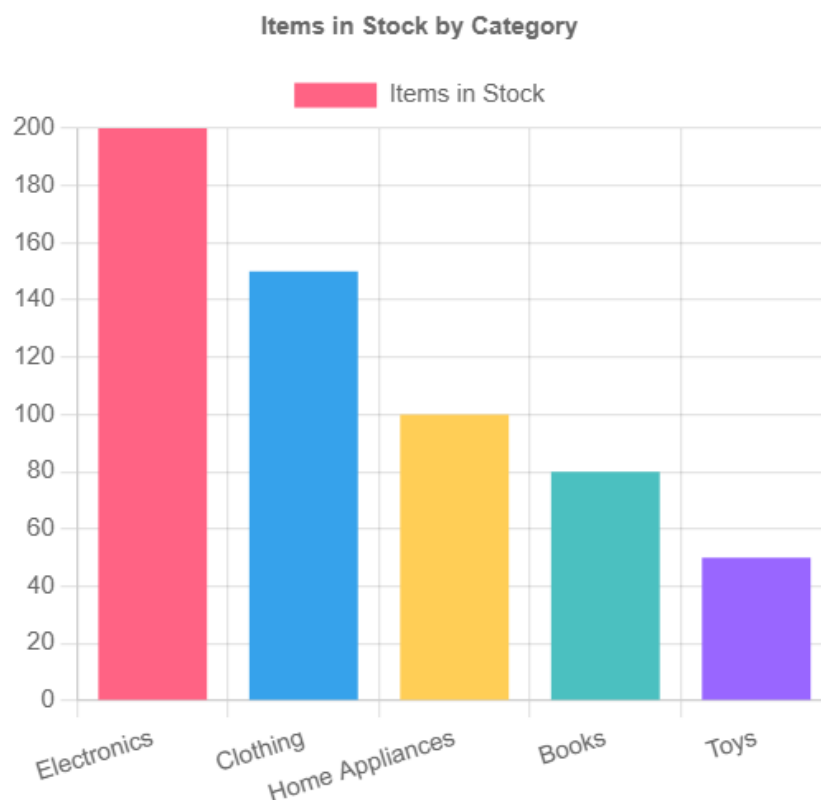
Chart Components:

Title: Inventory Management System

Legend: Shows the colour codes for each category: o Electronics (Pink) o Clothing (Blue) o Home Appliances (Yellow) o Books (Teal) o Toys (Purple)

Purpose: Helps in quickly identifying the dominant inventory category and overall allocation of stock. Insight Example: From the chart, Electronics and Clothing appear to occupy the largest sections, indicating these categories have the most inventory.

2. Items in Stock by Category – Bar Chart



The bar chart provides a detailed count of items available in stock for each product category. It is helpful for inventory tracking and restocking decisions.

Chart Components:

Title: Items in Stock by Category

Legend: Indicates that the bars represent the number of items in stock.

Axes:

X-axis: Categories – Electronics, Clothing, Home Appliances, Books, Toys.

Y-axis: Quantity of items (0 to 200 scale).

Purpose: Allows comparison between the stock levels of different categories.

Insight Example: The bar chart clearly shows: Electronics has the highest stock (200 items).

Toys has the lowest (50 items). This highlights areas of potential overstock or the need for replenishment.